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ARABIC OFFLINE HANDWRITTEN RECOGNITION CHARACTER BASED

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Introduction

- This work focuses on offline Arabic handwritten text recognition (HTR) at the character level.
- Most existing models are designed for Latin scripts and struggle with Arabic due to its cursive nature, contextual letter forms, and right-to-left direction.
- Current HTR systems are mainly word- or line-based, relying on encoder–decoder architectures and contextual decoding methods (e.g., Word Beam Search). These approaches are highly dependent on the training corpus and vocabulary, limiting their generalization ability.
- Despite strong results, they remain far from human performance in reading and transcribing the dataset images, especially with diverse writing styles, ambiguous characters, and spelling errors.
- **Motivation:** Humans primarily read handwriting by focusing on individual character shapes, not on context or spelling correction.
- **Our Approach:** We adopt a character-based recognition strategy to test this hypothesis. The goal is to improve robustness and generalization in offline Arabic HTR by emphasizing character-level visual understanding.

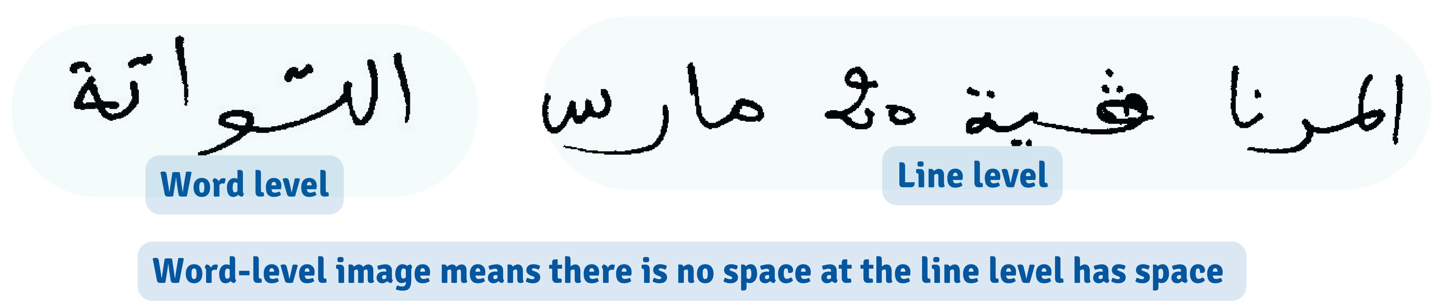

Fig 1: INF-ENIT word and line based examples

Methodology

The methodology consists of three stages: (1) preprocessing the INF-ENIT Arabic dataset by selecting word-level images and creating train/validation/test splits based on character distribution, (2) adapting and training multiple HTR approaches (encoder–decoder and object detection) with respect to their required annotation formats, input representations, and output structures, and (3) evaluating their performance using standard HTR metrics (Character Error Rate and Word Accuracy Rate) along with character swapping experiments to assess character-level generalization.

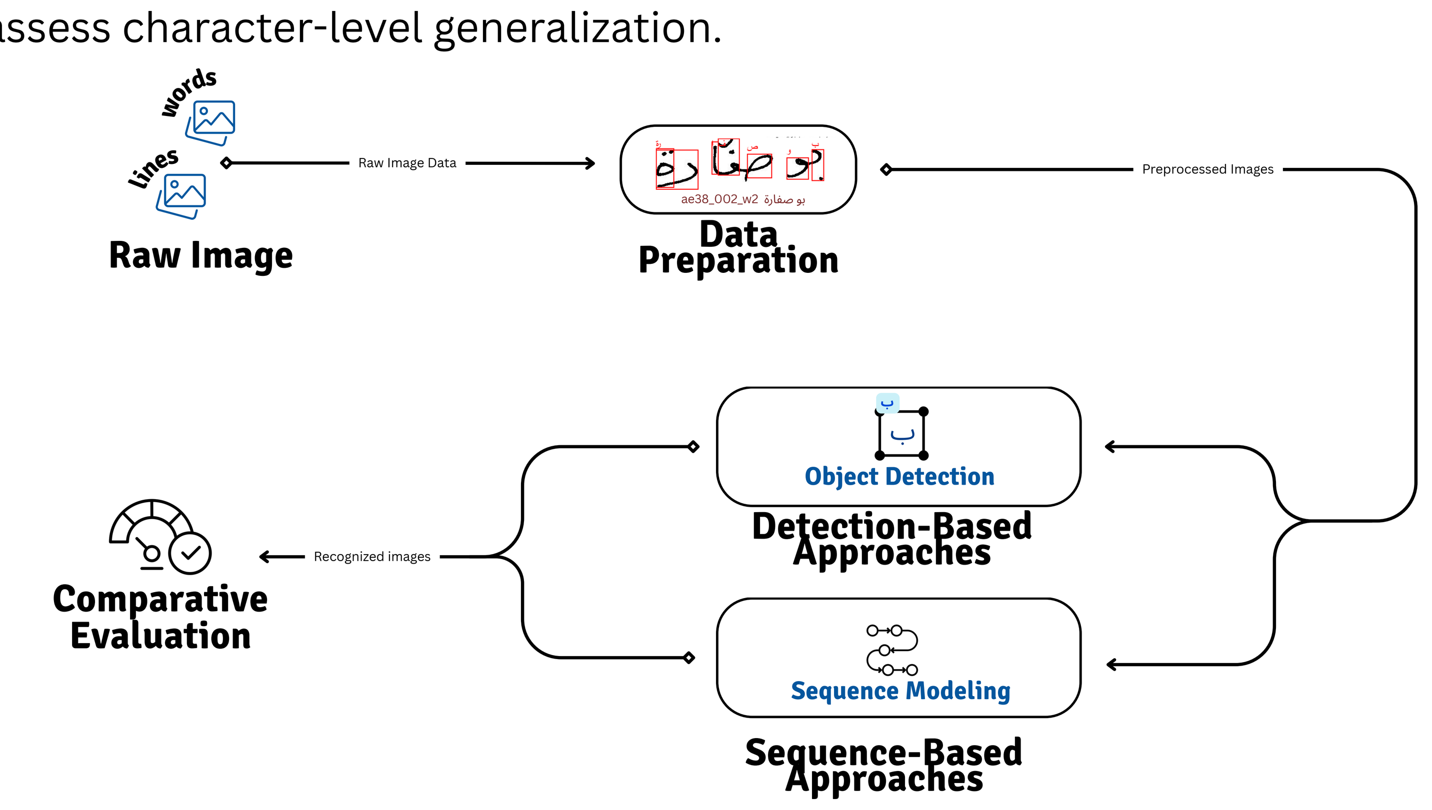

Fig 2: Our methodology workflow

Experiments

In the first stage, the INF-ENIT dataset is used as the main Arabic handwriting database. The dataset is first preprocessed to retain only word-level images by removing line-based samples. A statistical analysis of character distribution and word length is then performed to ensure a balanced dataset split. Based on this analysis, the data is divided into training, validation, and test sets. The annotations are further adapted to fit different offline HTR approaches, enabling consistent use across multiple model types.

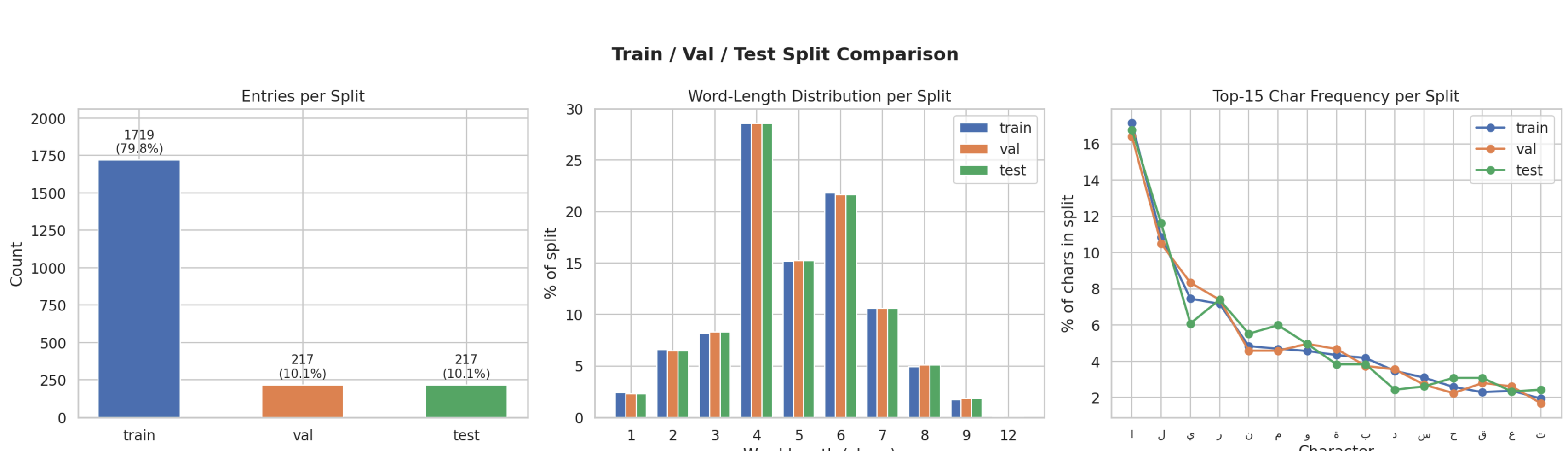

Fig 3: INF-ENIT dataset analysis and data split

In the second stage, the models are trained using a Faster R-CNN-based approach as a representative detection-based HTR method. Performance is evaluated using Character Error Rate (CER) and Word Accuracy Rate (WAR).

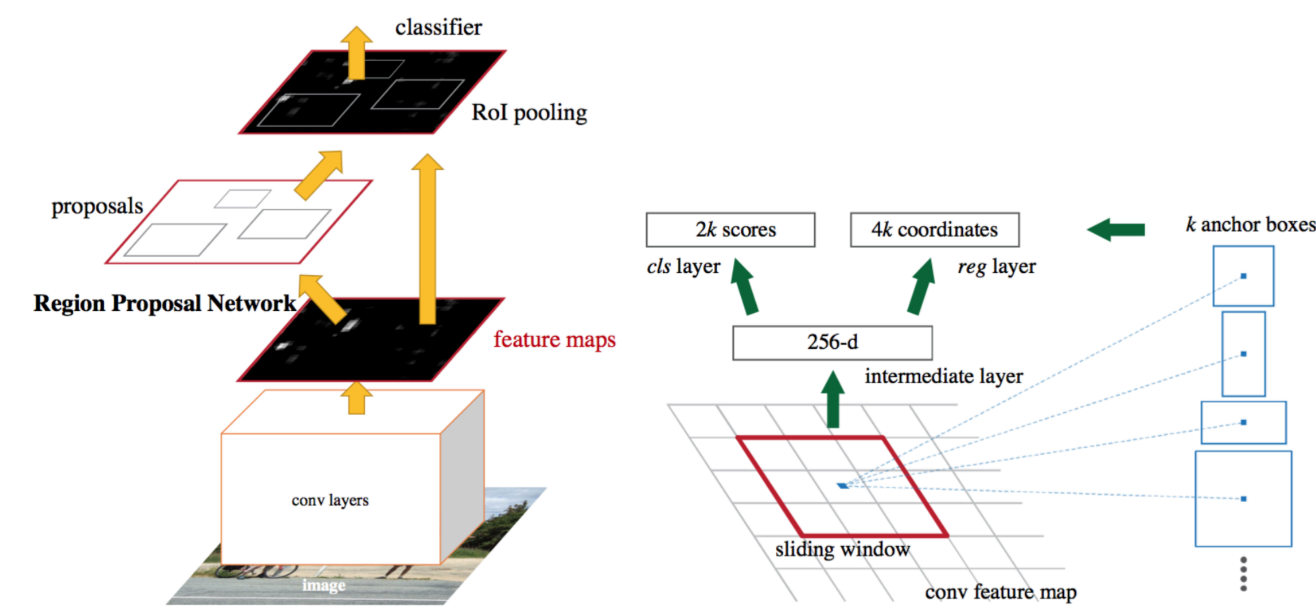

Fig 4: Faster Rcnn backbone model

Model	CER - Val	WAR - Val	CER - Test	WAR - Test
BLSTM-CTC BestPath	21.60%	60.18%	29.05%	52.60%
BLSTM-CTC WordBeamSea	**18.67%**	65.87%	**23.77%**	59.71%
Faster RCNN	27.67%	**70.37%**	29.77%	**74.19%**

Tab1: Character based vs Seq2seq based offline htr performance comparison

Character Error Rate (CER):

$$CER = \frac{\text{Number of incorrect characters}}{\text{Total number of characters in the reference text}} \times 100\%$$

Word Error Rate (WER):

$$WER = \frac{\text{Number of incorrect words}}{\text{Total number of words in the reference text}} \times 100\%$$

In addition, robustness is assessed through controlled character perturbation and swapping experiments. These tests are designed to measure the models' ability to generalize beyond contextual cues and rely on true character-level spatial understanding.

Fig 5: INF-ENIT Character perturbation examaple from training set

Model	CER Perturbation	WAR perturbation
BLSTM-CTC BestPath	79.68%	0.00%
BLSTM-CTC WordBeamSearch	79.68%	0.00%
Faster RCNN	**26.56%**	**70%**

Tab2: Character based vs Seq2seq based character perturbation performance comparison

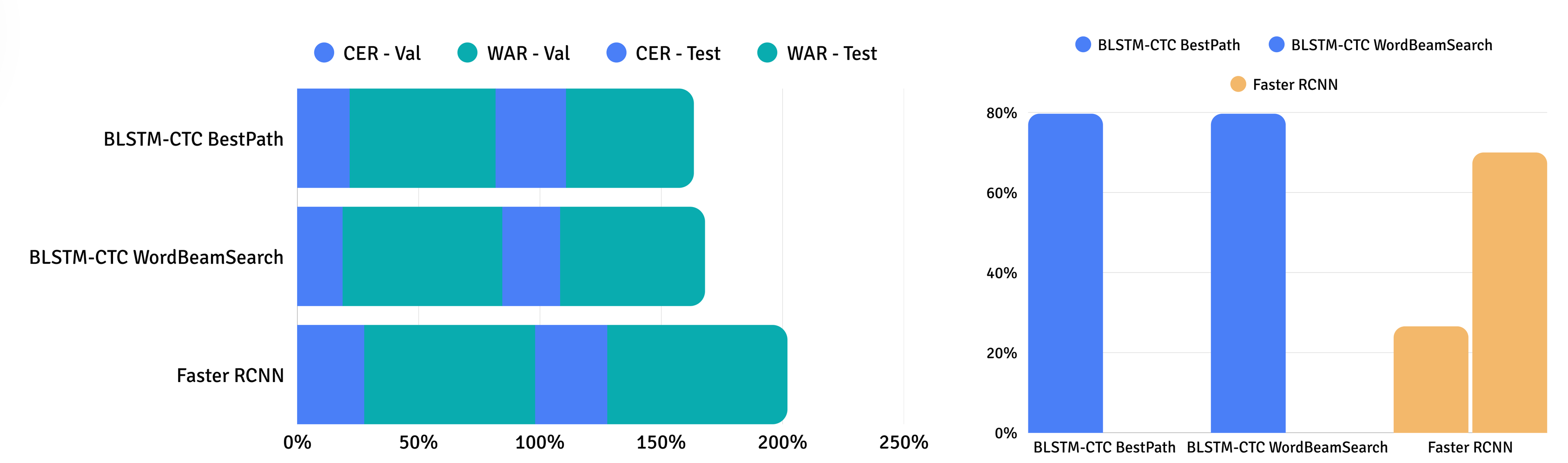

Fig 6: Experiments results analysis.

Conclusion

- Object detection approaches are closer to human behavior in handwritten text transcription.
- They show strong robustness to character perturbation, indicating better generalization than seq2seq models.
- On a balanced INF-ENIT dataset, Faster R-CNN achieves state-of-the-art performance.
- Future work will focus on data augmentation to address intrinsic and extrinsic handwriting variability.
- The final objective is to deploy this solution in real-world professional applications.